

Questions for the Audience

How to Connect Feature Map $\mathcal{A}$ to Fully-Connected Layer?

- Flattening: $\mathcal{A} \in \mathbb{R}^{Q \times K \times M} \rightarrow \vec{a} \in \mathbb{R}^{Q \cdot K \cdot M}$
- Global pooling over spatial dimensions $\mathcal{A} \in \mathbb{R}^{Q \times K \times M} \rightarrow \vec{a} \in \mathbb{R}^Q$
- Convolution with kernel size $K \times M$ of feature map

How many Jacobians do we Compute per Operation?

- In mathematical theory: Only a single Jacobian
- In practice: One per learnable **parameter** of the layer
- Plus one for all inputs of our function

Questions for the Audience

What are Reasons for Exploding Gradients

- Learning rate η is too large
- Weights in fully-connected and convolution layers are too large

What are Reasons for Vanishing Gradients

- Learning rate η is too small
- Weights in fully-connected and convolution layers are too small
- Derivative of sigmoid activation functions smaller than 1

Which Options for Transfer Learning Exist?

- Freeze all layers, remove last FC layer, use as feature extractor
- Possibly freeze some layers, replace last FC layer, fine-tune